

# Qianou Christina Ma

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## Research Focus

My research lies at the intersection of Human-AI Interaction, Generative AI, and the Learning Sciences. I **design, build, and evaluate LLM applications** to foster AI literacy, optimize human-AI task delegation, and support effective human-AI workflows, especially in **programming** and **computing education**. I conduct **mixed-methods research** to investigate how people collaborate and learn with AI systems, and design systems and frameworks to support training and evaluation of effective human-AI workflows.

## Education

### Ph.D. in Human-Computer Interaction

CARNEGIE MELLON UNIVERSITY (CMU) Advisors: DR. **KEN KOEDINGER** & DR. **SHERRY TONGSHUANG WU**

*Dissertation:* Training for the Future: Preparing Humans for Effective Human-AI Programming

*Thesis Committee:* Dr. Sherry Tongshuang Wu, Dr. Ken Koedinger, Dr. Brad Myers, Dr. Elena Glassman

Sept. 2022 - now

Pittsburgh, PA

### M.S. in Human-Computer Interaction

CARNEGIE MELLON UNIVERSITY (CMU)

*Relevant Coursework:* Deep Learning; Advanced NLP; LLM Systems; AI Tools for SDE; Learning Media Design; Persuasive Design

Dec. 2024

Pittsburgh, PA

### B.S. in Computer Science & B.S. in Cognitive Science

CARNEGIE MELLON UNIVERSITY (CMU) MINOR IN DESIGN FOR LEARNING. QPA: 3.91/4.0

May 2022

Pittsburgh, PA

## Research & Work Experience

### Applied Science Intern, Microsoft

MENTORS: DR. SHAMSI IQBAL, DR. SIDDARTH SURI & DR. PRANAV KHAPE

- Leading an *empirical study* (paper in submission [P1]) on how expertise shapes agentic work and translating research to AI upskilling insights.
- Analyzing *human profiles* and *interaction behaviors and trajectories* to identify factors that predict a better future of work with LLM agents.

Redmond, WA

2026 Summer

### PhD Researcher, Carnegie Mellon University

ADVISORS: DR. KEN KOEDINGER & DR. SHERRY TONGSHUANG WU

- Designed and built new measures and LLM-based learning tools to develop AI literacy resilient to model updates (see summary at [W1]). E.g., a teachable agent for CS1 debugging (HypoCompass [P12]) and a tutor to help end-users program in prompts (ROPE [P10]).
- Led *mixed-methods research* (literature reviews [W7, P11], controlled experiments [P10], classroom study [P8], and pre-post user studies [P12, P6], and conducted statistical modeling, log analysis, and qualitative coding to evaluate human-AI collaboration and learning effectiveness.
- Collaborated with *interdisciplinary teams* to co-author 10+ peer-reviewed papers (e.g., NLP [P5, W6], CSEd [P14, W5], SWE [P9], HCI [P8]), including a first-authored Best Paper [P12] and Honorable Mention [P13] and multiple invited presentations (IJCAI, CSCL).

Pittsburgh, PA

2022 - present

## Selected Publications

### MANUSCRIPTS AND PRE-PRINTS

- [P1] **Qianou Ma**, Pranav Khape, Rock Pang, Ken Koedinger, Tongshuang Wu, Siddarth Suri, and Shamsi Iqbal. Impact of expertise and trust on ai-assisted work. In *Submission*, 2027.
- [P2] Yeon Su Park, Nadia Arvi, Hae Ri Lee, Sehoon Lim, **Qianou Ma**, and Juho Kim. **The Work Behind Delegation: A Framework for Supervising AI Coding Agents**. In *Submission*, 2027.
- [P3] Zora Zhiruo Wang, Apurva Gandhi, Rulin Shao, Aspen Chen, Jonas Mueller, Zhiqi Liang, Jett Chen, Michael Ryan, **Qianou Ma**, Luxi He, et al. Efficient test-time adaptation through human-ai interaction. In *NeurIPS TTCL Workshop (accepted)*, 2026.
- [P4] Xuhui Zhou\*, Weiwei Sun\*, Weihua Du, Jiarui Liu, Haojia Sun, **Qianou Ma**, Tongshuang Wu, Yiming Yang, and Maarten Sap. **OdysSim: Building Foundation Models for Human Behavior Simulation**. In *NeurIPS (accepted)*, 2026. \*Equal contribution.

- [P5] Xuhui Zhou\*, Weiwei Sun\*, **Qianou Ma**, Yiqing Xie, Jiarui Liu, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Tongshuang Wu, et al. [Mind the Sim2Real Gap in User Simulation for Agentic Tasks](#). In *COLM (accepted)*, 2026. \*Equal contribution.

## PEER-REVIEWED CONFERENCE AND JOURNAL PUBLICATIONS

- [P6] **Qianou Ma**, Megan Chai, Yike Tan, Jihun Choi, Jini Kim, Erik Harpstead, Geoff Kauffman, and Tongshuang Wu. [“GenAI Defaults to Bias!” Gamify AI literacy through Reflections on Prompts](#). In *International Conference on Artificial Intelligence in Education (AIED)*, 2026. Code: </>.
- [P7] **Qianou Ma\***, Keyu He\*, Valerie Chen, Wayne Chi, and Tongshuang Wu. [RECAP: An End-to-End Platform for Capturing, Replaying, and Analyzing AI-Assisted Programming Interactions](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 2026. Code: </>. \*Equal contribution.
- [P8] **Qianou Ma**, Ken Koedinger, and Tongshuang Wu. [Not Everyone Wins with LLMs: Behavioral Patterns and Pedagogical Implications for AI Literacy in Programmatic Data Science](#). In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. Code: </>.
- [P9] Chenyang Yang, Yike Shi, **Qianou Ma**, Michael Xieyang Liu, Christian Kästner, and Tongshuang Wu. [What Prompts Don’t Say: Understanding and Managing Underspecification in LLM Prompts](#). In *Findings of the Association for Computational Linguistics: ACL*, 2026.
- [P10] **Qianou Ma**, Weirui Peng, Chenyang Yang, Hua Shen, Kenneth Koedinger, and Tongshuang Wu. [What Should We Engineer in Prompts? Training Humans in Requirement-Driven LLM Use](#). In *Transactions on Computer-Human Interaction (TOCHI)*, 2025. Presented at CHI’26. Media coverage: [SCS News](#); [New York Times](#). Code: </>.
- [P11] **Qianou Ma\***, Dora Zhao\*, Xinran Zhao, Chenglei Si, Chenyang Yang, Ryan Louie, Ehud Reiter, Diyi Yang, and Tongshuang Wu. [SPHERE: An Evaluation Card for Human-AI System Evaluation](#). In *ACL Findings*, 2025. Media coverage: [MIT AHA Workshop](#). Code: </>. \*Equal contribution.
- [P12] **Qianou Ma**, Hua Shen, Kenneth Koedinger, and Tongshuang Wu. [How to Teach Programming in the AI Era? Using LLMs as a Teachable Agent for Debugging](#). In *International Conference on Artificial Intelligence in Education (AIED)*, 2024. 🏆 **Best Paper** 🏆 **Best Interactive Event**. Invited presentations at IJCAI’25 and CSCL’25. Media coverage: [The Link](#), [New York Times](#). Code: </>.
- [P13] Atharva Naik, Jessica Ruhan Yin, Anusha Kamath, **Qianou Ma**, Sherry Tongshuang Wu, Charles Murray, Christopher Bogart, Majd Sakr, and Carolyn P. Rose. [Generating Situated Reflection Triggers About Alternative Solution Paths: A Case Study in Generative AI for Computer-Supported Collaborative Learning](#). In *International Conference on Artificial Intelligence in Education*. Springer, 2024. 🏆 **Best Paper Nominee**.
- [P14] Atharva Naik, Jessica Ruhan Yin, Anusha Kamath, **Qianou Ma**, Sherry Tongshuang Wu, R. Charles Murray, Christopher Bogart, Majd Sakr, and Carolyn P. Rose. [Providing tailored reflection instructions in collaborative learning using large language models](#). *British Journal of Educational Technology*, 20, 2024.

## PEER-REVIEWED WORKSHOP, POSTERS & WORK-IN-PROGRESS

- [W1] **Qianou Ma**. [Training for the Future: Preparing Humans for Effective Human-AI Programming](#). In *CHI EA’26: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (Doctoral Consortium)*, 2026.
- [W2] **Qianou Ma**, Sherry Tongshuang Wu, and Kenneth R. Koedinger. [What Calculus Should We Teach in the Age of AI? Identifying and Improving Higher-Order Thinking in College Math](#). In *Artificial Intelligence in Education Doctoral Consortium*. 2026.
- [W3] Ruiwei Xiao, Shan Zhang, Xinying Hou, Ying-Jui Tseng, **Qianou Ma**, Yash Tadimalla, Qing Xiao, Jionghao Lin, Bo Jiang, John Stamper, and Kenneth R. Koedinger. [AI Literacy For All: 2nd International Workshop on AI Literacy Education For All](#). In *Artificial Intelligence in Education FoL Workshops*. 2026.
- [W4] **Qianou Ma**, Anika Jain\*, Jini Kim\*, Megan Chai\*, and Geoff Kauffman. [ImaginAltion: Promoting Generative AI Literacy Through Game-Based Learning](#). In *CHI EA’25: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2025. \*Equal contribution.
- [W5] Aysa Xuemo Fan, Rully Agus Hendrawan, Yang Shi, and **Qianou Ma**. [Enhancing Code Tracing Question Generation with Refined Prompts in Large Language Models](#). *SIGCSE TS (55th ACM Technical Symposium on Computer Science Education)*, 2024.
- [W6] **Qianou Ma\***, Yann Hicke\*, Anmol Agarwal\*, and Paul Denny. [AI-TA: Towards an Intelligent Question-Answer Teaching Assistant using Open-Source LLMs](#). In *Proceedings of the Workshop on GenAI for Education @NeurIPS*, 2023. \*Equal contribution.
- [W7] **Qianou Ma**, Tongshuang Wu, and Kenneth Koedinger. [Is AI the better programming partner? Human-Human Pair Programming vs. Human-AI pAIr Programming](#). In *Proceedings of the Workshop on LLMs@AIED*, 2023. Presented at ICER’23.

## Awards & Fellowships

### HONORS & AWARDS

|      |                                                                                                   |                |
|------|---------------------------------------------------------------------------------------------------|----------------|
| 2026 | <b>Fellow on AI &amp; Society</b> , Block Center for Technology and Society                       | Pittsburgh, PA |
| 2024 | <b>Best Paper</b> , top-1 best paper [P12] awarded at AIED 2024                                   | Recife, Brazil |
| 2024 | <b>Best Interactive Event</b> , top-1 best interactive event [P12] at AIED 2024                   | Recife, Brazil |
| 2024 | <b>Best Paper Nominee</b> , top-6 best paper nominee [P13] at AIED 2024                           | Recife, Brazil |
| 2023 | <b>Generative AI Innovation Incubator (GAI3) Hackathon</b> , 1st place team awarded with \$20,000 | Pittsburgh, PA |
| 2021 | <b>Phi Beta Kappa</b> , 1 of 16 seniors selected in Fall 2021                                     | Pittsburgh, PA |

### FELLOWSHIPS & FUNDINGS

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|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 2026 | <b>What Calculus Should We Teach in the Age of AI? Identifying and Improving Higher-Order Thinking in Math</b> , Learnvia Seed Grant (Role: Co-PI; led proposal writing) | \$140,000 |
| 2025 | <b>RealEduArena: Evaluating LLMs' Pedagogical Implementations in Real AI Tutoring</b> , Google Gift Fund on Evaluations and Benchmarks (Role: lead PhD researcher)       | \$75,000  |
| 2025 | <b>ImaginAltion: Promoting Generative AI Literacy Through Game-Based Learning</b> , CMU GSA/Provost Conference Fund                                                      | \$1500    |
| 2024 | <b>Explicitly Train Humans towards Human-AI Collaboration</b> , Google Academic Research Award (Role: lead PhD researcher; contributed to proposal writing)              | \$100,000 |
| 2021 | <b>Support Designer-Teacher Collaboration in Educational Game Design Using Learning Science Principles</b> , CMU Dietrich College Senior Honors Fellowship               | \$6500    |
| 2020 | <b>The contribution of local and global language statistics to developing structured semantic networks</b> , CMU Summer Undergraduate Research Fellowship                | \$3500    |
| 2020 | <b>Improving Web Visualization to Support Teachers &amp; Parents: a Case Study of the Carnegie Museum of Art</b> , CMU Small Undergraduate Research Grant                | \$500     |
| 2019 | <b>Linguistic patterns differentially predict children's semantic structure relatedness</b> , Ireland Undergraduate Research Fund                                        | \$1750    |

## Presentations & Talks

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|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 2026 Summer | <b>Invited Talk</b> , The Impact of Expertise on AI-assisted Work                                                                                              | Microsoft               |
| 2026 Spring | <b>Invited Talk</b> , What to Teach and How to Teach Programming in the AI Era<br>Penn; Cornell; Columbia; UMich; UChicago; UIUC; UCI; UCSD; UCB; Stanford; UW | Various Schools         |
| 2025 Fall   | <b>Invited Talk</b> , Training Future Proof Developers                                                                                                         | HKU; HKUST; ECNU        |
| 2025 Fall   | <b>Invited Keynote</b> , AI x Future of Work Series: Workforce Preparation on AI Literacy                                                                      | Plug & Play Tech Center |
| 2025 Fall   | <b>Invited Talk</b> , Training Requirement-Driven LLM Use for Prompt Programming                                                                               | Mila HCAI seminar       |
| 2025        | <b>Invited Talk</b> , Training the Use of LLMs on Programmatic Data Science for Non-Experts                                                                    | Bloomberg; Pitt         |
| 2025 Summer | <b>Invited Talk</b> , IJCAI 2025 Sister Conferences' Best Paper Presentation                                                                                   | Montreal, Canada        |
| 2025 Summer | <b>Invited Talk</b> , CSCL/ISLS 2025 IAALDE-Network Best Paper Presentation                                                                                    | Helsinki, Finland       |
| 2024 Fall   | <b>Invited Talk</b> , Scaling Student Support: Online Question-Answering with LLM Tutors                                                                       | Google DeepMind         |
| 2024 Fall   | <b>Invited Talk</b> , Training for the Future: Preparing Students for Human-LLM Partnerships                                                                   | UMich; CMU; Pitt        |

## Teaching & Mentoring Experience

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|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2023-present | <b>Mentor</b> , Independent Studies or Projects on LLM-assisted Learning<br>Advised students from LS, HCI, CS, and NLP on various projects [P10, P13, P12, W4], now UMich PhD, CMU PhD, etc. | 8 Undergrad/grad; 1 HS |
| 2024 Fall    | <b>Mentor</b> , CMU Paths to AI Research Program for traditionally underrepresented                                                                                                          | 2 Undergraduates       |
| 2022, 2024   | <b>Mentor</b> , CMU SCS Graduate Application Support Program for underrepresented                                                                                                            | 3 Master's students    |
| 2023, 2024   | <b>Mentor</b> , CMU LearnLab Summer School Computing Education Research (e.g., [W5])                                                                                                         | 7 Ph.D. and 1 Master's |

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|-------------|-------------------------------------------------------------------------------------------------|---------------------|
| 2026 Fall   | <b>Guest Lecturer</b> , CMU 05-315/615 Persuasive Design                                        | 60+ Undergrad/grad  |
| 2026 Spring | <b>Guest Lecturer</b> , CMU 05-840 Tools for Online Learning                                    | 25+ Undergrad/grad  |
| 2026 Spring | <b>Guest Lecturer</b> , CMU 05-499/899 AI Augmented Designer                                    | 60+ Undergrad/grad  |
| 2025 Fall   | <b>Guest Lecturer</b> , University of Michigan EECS-498 AI in Education                         | 15+ Undergrad/grad  |
| 2025 Summer | <b>Guest Lecturer</b> , Northeastern INFO-7375 Prompt Engineering for Generative AI             | 40+ Undergrad/grad  |
| 2025 Spring | <b>Guest Lecturer</b> , CMU 05-898 HCI & Data Science for Product Managers                      | 40+ Undergrad/grad  |
| 2024 Summer | <b>Guest Lecturer</b> , CMU 82-183 AI for Humanities: The Multi-dimensions of World             | 20+ Undergraduate   |
| 2024 Fall   | <b>Teaching Assistant</b> , CMU 05-430/630 Programming Usable Interfaces                        | 70+ Undergrad/grad  |
| 2023 Fall   | <b>Teaching Assistant</b> , CMU 05-391/891 Design Human-Centered Software                       | 60+ Undergrad/grad  |
| 2021 Fall   | <b>Teaching Assistant</b> , CMU 15-110 Principles of Computing                                  | 300+ Undergraduates |
| 2021 Summer | <b>Teaching Assistant</b> , CMU CS Scholar Pre-College Program <i>for underrepresented</i>      | 30+ Grades 9-12     |
| 2021 Spring | <b>Teaching Assistant</b> , CMU <a href="#">15-150 Functional Programming (Reflection Blog)</a> | 300+ Undergraduates |

## Service & Leadership Experience

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|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2024 - present | <b>Program Committee</b> , Various conferences<br>NeurIPS GAIED (2023), EDM (2024-26), SIGCSE (2025), ICLR BiAlign (2025), AIED (2026), ALIT4ALL (2026), NeurIPS BiAlign (2026)                                                                                                                                                            |                    |
| 2022 - present | <b>Reviewer</b> , Various conferences and journals<br>ACM CHI (2023-26), UIST (2024-26), Computers & Education (2024)<br>🏆 <i>Special Recognitions for Outstanding Reviews</i> (ACM CHI)                                                                                                                                                   |                    |
| 2019 - present | <b>Member</b> , CMU University Disciplinary Committee & Academic Review Board                                                                                                                                                                                                                                                              | Pittsburgh, PA     |
| 2026 - 2027    | <b>Publicity Chair</b> , Human-AI Interaction Conference (HAIC)                                                                                                                                                                                                                                                                            | Washington, DC     |
| 2026 Summer    | <b>Student Volunteer</b> , CHI (2026)                                                                                                                                                                                                                                                                                                      | Barcelona, Spain   |
| 2026 Summer    | <b>Organizer</b> , AIED Workshop: 2nd Workshop on AI Literacy Education for All <a href="#">[W3]</a>                                                                                                                                                                                                                                       | Seoul, South Korea |
| 2026 Spring    | <b>Panelist</b> , CHI Workshop: Human-Human Collaboration to Human-Agent Collaboration                                                                                                                                                                                                                                                     | Barcelona, Spain   |
| 2025 Fall      | <b>Invited Expert</b> , MIT Media Lab Benchmarks for Human Flourishing with AI Workshop                                                                                                                                                                                                                                                    | Boston, MA         |
| 2025 Spring    | <b>Co-leader</b> , NSF I-Corps Team for LLM-powered Teaching Assistant Project                                                                                                                                                                                                                                                             | Pittsburgh, PA     |
| 2025 Spring    | <b>Member</b> , HCII NSF Research Experiences for Undergrad (REU) Admissions Committee                                                                                                                                                                                                                                                     | Pittsburgh, PA     |
| 2025 Spring    | <b>Member</b> , CMU HCII Ph.D. Admissions Committee                                                                                                                                                                                                                                                                                        | Pittsburgh, PA     |
| 2025 Spring    | <b>Panelist</b> , K-12 AI Teacher Professional Development at Eberly Center                                                                                                                                                                                                                                                                | Pittsburgh, PA     |
| 2024 Fall      | <b>Panelist</b> , CMU School of Computer Science Freshman Immigration Course                                                                                                                                                                                                                                                               | Pittsburgh, PA     |
| 2023 Spring    | <b>Director</b> , CMU School of Computer Science Graduate Student Musical<br>Organize and direct a musical revue with 30+ performers and 200+ audience                                                                                                                                                                                     | Pittsburgh, PA     |
| 2022 Fall      | <b>Co-leader</b> , CMU Women@SCS TechNights (data bias session)<br>Help organize a session on training data and a data bias card game for 20+ middle-school Black girls                                                                                                                                                                    | Pittsburgh, PA     |
| 2021 Fall      | <b>Member</b> , CMU Teaching Innovation Award Selection Committee                                                                                                                                                                                                                                                                          | Pittsburgh, PA     |
| 2018 - 2022    | <b>President (21-22), VP of Communications (20), Project Advisor (18-19), Project Ignite</b><br>Recruit college students to provide free interdisciplinary project-based learning opportunities to local high-school students, increase diversity in high-school student applications, and facilitate about 100 workshop sessions per year | Pittsburgh, PA     |
| 2018 - present | <b>Docent</b> , Carnegie Museum of Art ( <a href="#">CMOA</a> ), leading public tours and school tours                                                                                                                                                                                                                                     | Pittsburgh, PA     |